

**Mini Project Report
On
CROWDFUNDING ON BLOCKCHAIN TECHNOLOGY**

Submitted By

**0105 Kajal Barewar
0109 Krusha Patel
0131 Durgesh Khadgade
0140 Prabhat Ranasingh
0144 Prajyot Waikar**

Guided by

DR.P.S.PRASAD



**DEPARTMENT OF INFORMATION TECHNOLOGY
PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR
SESSION 2021-22**

**DEPARTMENT OF INFORMATION TECHNOLOGY
PRIYADARSHINI COLLEGE OF ENGINEERING
NAGPUR-440019**

Certificate

This is to certify that the B-Tech Mini project entitled

CROWDFUNDING ON BLOCKCHAIN TECHNOLOGY

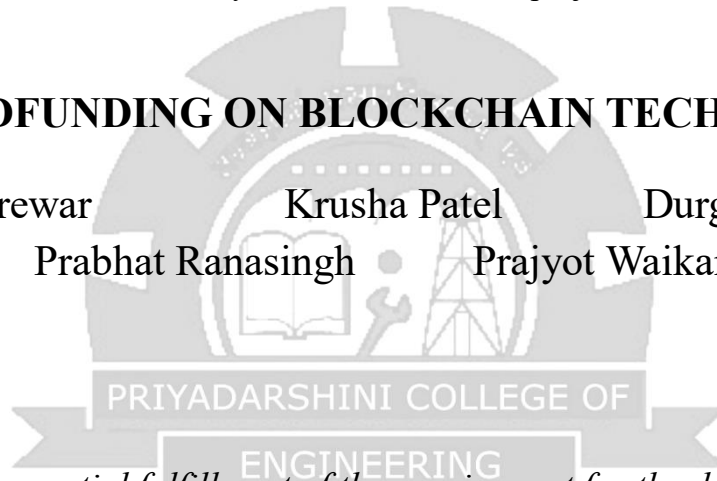
Kajal Barewar

Krusha Patel

Durgesh Khadgade

Prabhat Ranasingh

Prajyot Waikar



*In the partial fulfillment of the requirement for the degree of
‘Bachelor Of Technology’ in Information Technology of
Priyadarshini College of Engineering, Nagpur is bonafide work carried under
guidance and supervision*

Dr.P.S. Prasad
Project Guide

Dr. P. M. Chaudhari
H.O.D. (IT)

Dr. S. A. Dhale
Principal
P.C.E., Nagpur

Acknowledgement

We have a great honor to acknowledge Dr. S. A. Dhale, Principal, Priyadarshini College of Engineering, Rashtrasant Tukdoji Maharaj Nagpur University, and Nagpur, who had given us his consent to carry out this project.

We are very much delighted to voice my gratitude and indebtedness to Dr. P. M. Chaudhari, Head of Department of Information Technology, Priyadarshini College of Engineering, Rashtrasant Tukdoji Maharaj Nagpur University, Nagpur, who had given us his consent to carry out this Project.

We feel immense pleasure and privilege in expressing my deep sense of gratitude towards my Mini Project Guide, whose valuable guidance and critical analysis of my result has led to successful completion of our project.

Our special thanks to all my Friends for giving me incentive support in this report work.

We express our indebtedness and our gratitude to our affectionate and loving family and friend for encouragement and enthusiastic support throughout this study.

We thank to our respected parents, whose patience and support was instrumental in accomplishing this task.

Kajal Barewar
Krusha Patel
Durgesh Khadgade
Prabhat Ranasingh
Prajyot Waikar

INDEX

SR.NO	TITLE	PAGE NO.
1.	Abstract	05
2.	Introduction	06-07
3.	Literature Review	08-10
4.	Research Methodology	11
5.	Flowchart	12
6.	Problem Definition	13
7.	Requirement Specification	14
8.	System Requirement	15
9.	Application	16
10.	Future Enhancement	17
11.	Conclusion	18
12.	Reference	19

ABSTRACT

Many organizations are raising money to aid local governments in locating additional funding sources that will be used to provide aid to those in need in the Covid-19 plague situation. For the parties engaged in generating these funds, including the donor, the service provider of the fundraising platform, and even the fundraiser, trust is a crucial component. This time, the author attempts to analysis how to integrate smart contracts and blockchain technology into the prevalent crowdfunding process schemes. The findings of this study suggest that blockchain-based smart contracts can be used to implement the most popular crowdfunding process schemes.

Blockchain-based crowdfunding enables peer-to-peer fundraising without the need for intermediaries. It is a decentralized funding mechanism. For investors to make financial contributions and for project creators to obtain funding, the blockchain network serves as a transparent and secure platform.

Smart contracts allow for the automatic execution of crowdfunding campaigns with specified rules and conditions, guaranteeing that funds are only distributed after certain requirements have been satisfied. This feature increases party confidence and openness while lowering the likelihood of fraud.

Additionally, blockchain platforms that are based on crowdfunding can provide a variety of advantages, including lower transaction costs, quicker settlement times, and greater accessibility. Additionally, these platforms may make it easier for smaller investors to take advantage of investment opportunities that were formerly only available to institutional investors.

In general, crowdfunding-based blockchain has the ability to completely change the fundraising industry by offering a faster, more open, and more accessible method of gathering money for projects.

INTRODUCTION

Nearly all industrial sectors in this fourth industrial revolution (I4.0) use technology to help their operations, whether they are for-profit or non-profit organization Non-profit organizations (non-profit) and profit-oriented organizations (profit) almost share the same qualities, however there are differences in the issues these two organizations face.

Nearly all nations currently dealing with the Covid-19 plague face the same challenges, particularly in terms of the resources required. The government has employed a variety of measures to manage its resources in order to combat the Covid-19 disease epidemic. The community also decided to gather money in response to this circumstance to aid the government in combating the Covid-19 plague outbreak.

Of course, raising money is a difficult process since it involves building trust between numerous parties, including funders, intermediaries, and organizations that act as a repository for temporary funds for recipients. This trust serves as the primary source of funding for organizations that organize charitable giving, luring donors to contribute money to those in need.

Many non-profit organizations participate in fundraising efforts, particularly in light of the Covid-19 plague. Gaining contributors' trust is their biggest obstacle to raising money for the organization. Many non-profit organizations use technology to make it simple for donors to make financial contributions to them. Therefore, it can be inferred from this that technology also plays a significant role in obtaining as many funds as possible, in addition to trust, which is the key factor.

Based on this, the author attempts to analysis the procedures that are typically used in this fundraising organization using blockchain technology, which can be a substitute solution to raise funders' trust, which will undoubtedly have an impact on the amount of money the fundraising organization will be able to raise. If all requirements are satisfied, recipients can easily receive these funds thanks to smart contract technology. According to earlier studies, this technique can be applied to the telecommunications and medical fields.

The adoption of blockchain technology not only improves fund-raising organizations' credibility, but it can also be used as proof from funders that contributions are coming from trustworthy sources and that beneficiaries are who they claim to be.

1.1 Purpose

Various crowdfunding platforms exist, and they act as intermediaries between investors and fundraisers. These don't come without limitations, though. The introduction of Blockchain into crowdfunding helps improve the operations of crowdfunding platforms by providing the following advantages.

1. Through smart contracts, the costs of operation are reduced, and fundraisers are saved the portion of the money they would have had to pay to the crowdfunding platform from the raised funds.
2. Blockchain crowdfunding provides a high level of security and prevention of scam projects.
3. Most crowdfunding platforms focus on fundraising for tangible products, but with Blockchain involvement in crowdfunding, more innovative projects and services can be funded.
4. Increase in the accessibility of fundraising opportunities in different regions of the world.
5. They provide instant notification to members on the platform about new startups that can be invested in, so these startups need not pay for marketing and advertising, which would increase costs on limited funds.
6. Protection of intellectual property, thus reducing the risk of theft of ideas by investors who have the resources.
7. Faster transactions and cheaper transaction fees.

1.2 Scope

Innovations are changing everything around us, and blockchain technology is not an exception. About 0.5% of the world's population is using blockchain technology nowadays, and that's about 40 million people.

Since this technology is secure, fast, and transparent, it can be used to enhance businesses. And since it's widely used in crowdfunding app development, blockchain is officially trending in this domain. Even Kickstarter recently announced that they're moving to decentralized technology.

LITERATURE REVIEWS

2.1 Crowdfunding

Crowdfunding is a method of connecting between entrepreneurs and investors to invest in small amounts with an internet-based platform. Crowdfunding is divided into three categories according to the funding base offered by the crowdfunding platform.

The three categories are:

1. Donate
2. Pre-selling
3. Equity Crowdfunding

Entrepreneurs, crowdfunding platforms, and investors make up the key selection factors in crowdfunding. Each of the major stakeholders has a specific role and interest.

Entrepreneurs (companies or startups) present ideas, make funding requests using platforms for crowdsourcing, and then guarantee returns to investors in the first flow. Backers (investors) will examine the investment potential provided by entrepreneurs before committing to provide funding or guidance. It takes a platform that serves as a mediator to bring investors and supporters together.

2.2 Smart Contract

A smart contract is a contract that can be automatically performed and must be carried out in accordance with the terms of the contract. While some components of a smart contract can run automatically on a computer, human input and direction. can be implemented or executed through computer code that cannot be modified or approved, as well as through the legal enforcement of rights. Blockchain technology can be used in conjunction with smart contracts to automate complex transactions.

To enable, carry out, and enforce the conditions of a contract or agreement, a smart contract is a piece of code that may be performed on a blockchain network. After a contract's terms have been met, the main goal of a smart contract is to fulfil the agreement.

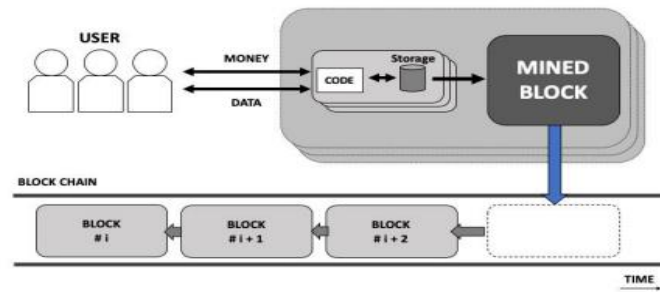


Figure 1: Smart Contract System [20]

A smart contract's account balance, personal storage, and executable code are represented in figure 1. Every time the smart contract is invoked, one associated block that stores the status of the contract is updated. After the contract is entered, the blockchain code cannot be altered. By submitting a transaction to a special address located 20 bytes from the contract, the contract can be performed. The contract is then carried out by the network's miners to reach consensus output, after which its status will be renewed. Contracts can create new contracts, read or write transactions to their personal storage, save money to their accounts, and receive messages or money from other users.

Smart contracts come in deterministic and nondeterministic varieties. A deterministic smart contract is one that, upon execution, doesn't need data from outside sources (sources not connected to the blockchain). Contracts that depend on information are known as non-deterministic contracts; examples of these include those that rely on data feeds or Oracle. One example of information that is not present on the blockchain is a contract to obtain weather information.

2.3 Blockchain

The major goal of the blockchain is to solve the growing issues, particularly those related to trust.

Blockchains are digital ledgers that can withstand distribution-mode destruction. Blockchain is a decentralized database that contains a general ledger of all completed transactions that is shared with all participants and is validated by a majority of the system's consensus. Additionally, data cannot be deleted once it has been entered. Every transaction that has ever taken place may be validated on the blockchain since it contains specific records. Blockchain is a database of transactional data that is distributed, verified, and maintained by global computer networks.

1. Header

contains information about a special reference number, the moment the block was produced, and a link to the block before it.

2. Content

The content includes a list of digital assets as well as the number of instructions that have been verified (transactions, addresses of those transactions, and the number of transactions).

The construction of a blockchain can also be illustrated in picture 5 below :

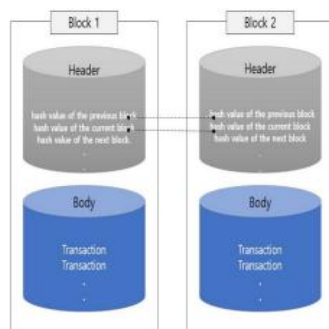


Figure 5: Structure of Blockchain [14]

Four key properties of blockchain exist:

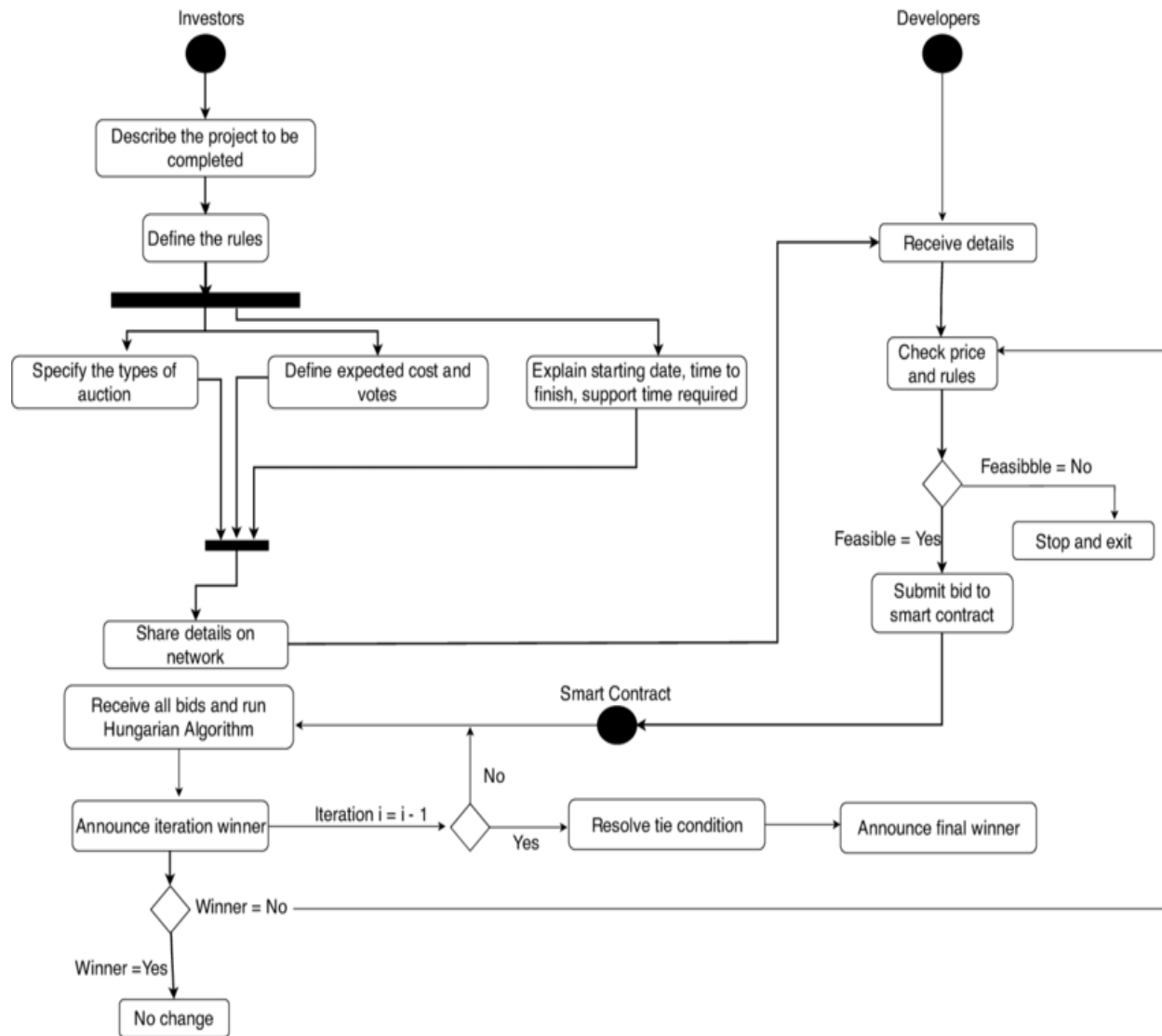
1. Immutable (permanent and tamper-proof) blockchain is a record of a transaction that cannot be altered, hence enhancing the confidence of the transaction record.
2. Decentralized: Every blockchain is saved in a file that can be read by any network (network copies).
3. Consensus Driven - (Trust Verification) Blockchain is independently validated using certain criteria that utilise special resources to demonstrate proof of the efforts performed.
4. Transparent - (full transaction history) all transaction data is available for all parties to access and audit.

RESEARCH METHODOLOGY

A literature review, literature study, theoretical study, theoretical foundation, and literature study are all synonyms for one another. The term "library research" refers to research that is only reliant on written works, encompassing both published and unpublished research findings.

According to the justification provided, the approach employed in this study is that of library research. In order for them to be used as references from this writing, this writing relies on sources from journals that have been published both worldwide and domestically as well as other pertinent works.

Flowchart



PROBLEM DEFINITION

While using blockchain technology for crowdfunding has many advantages, there are still a number of issues that need to be resolved. The absence of legal and regulatory requirements, which can be unsettling for investors and project developers, is one of the main problems.

Additionally, because of the nature of blockchain technology, security flaws and cyberattacks may affect blockchain-based crowdfunding campaigns. The smart contracts and wallets used in crowdfunding campaigns can be vulnerable to hacking even though blockchain technology is fundamentally safe.

Scalability is another problem with blockchain-based crowdfunding. The number of transactions that can be performed concurrently by the present blockchain infrastructure is constrained, which can result in lengthy transaction delays and expensive fees during times of heavy network traffic.

A barrier to entry for many investors and project creators unfamiliar with blockchain technology is the requirement for a high level of technical knowledge and skill for crowdfunding campaigns on the blockchain.

Finally, investors may find it challenging to assess the potential risks and returns of investment opportunities due to the absence of centralized intermediaries in blockchain-based crowdfunding campaigns. Investors may be put at a larger risk of losing their money as a result, which may deter them from taking part in crowdfunding initiatives.

Overall, even though blockchain-based crowdfunding has the potential to completely change the fundraising industry, these issues must be resolved to ensure the long-term adoption of the technology.

REQUIREMENT SPECIFICATIONS

- **Remix Ethereum IDE**
- **Solidity**

1. **Remix Ethereum IDE-** Remix IDE is an open source web and desktop application. It fosters a fast development cycle and has a rich set of plugins with intuitive GUIs. Remix is used for the entire journey of contract development as well as act as a playground for learning and teaching Ethereum.
2. **Solidity-** Solidity is an object-oriented, high-level language for implementing smart contracts. Smart contracts are programs which govern the behaviour of accounts within the Ethereum state. Solidity is a curly-bracket language designed to target the Ethereum Virtual Machine (EVM). It is influenced by C++, Python and JavaScript. You can find more details about which languages Solidity has been inspired by in the language influences section. Solidity is statically typed, supports inheritance, libraries and complex user-defined types among other features. With Solidity you can create contracts for uses such as voting, crowdfunding, blind auctions, and multi-signature wallets.

SYSTEM REQUIRED

System requirements are what is necessary for a client to install the web application in their system and be used hopefully without any difficulties. The aim of this is to help clients make sure they have all required tools or equipment. With analysis there are some objectives for hardware, software and any other thing that would be the minimum requirements to install the system. The minimum requirements are as follows.

- **PHP** – 7.3 (or higher recommended)
- **Database** – MariaDB – 10.1 or higher / MySQL – 5.7 or higher
- **WordPress 5.4 or higher**
- **Browser** – Chrome, Firefox, Safari (recommended)
Internet Explorer is not supported
- **Server Modules** – mod_rewrite, cURL, fsockopen, imagick

APPLICATION

Crowdfunding on blockchain technology has gained popularity in recent years due to its potential to create a transparent and secure platform for fundraising. Here are some applications of crowdfunding on blockchain technology:

1. **Decentralized Crowdfunding Platforms:** Blockchain technology allows for the creation of decentralized crowdfunding platforms, which eliminates the need for intermediaries like banks or crowdfunding websites. This can reduce transaction fees and increase transparency and trust between the fundraisers and investors.
2. **Smart Contracts:** Blockchain technology enables the use of smart contracts, which can automate the process of crowdfunding, eliminate the need for intermediaries, and provide security for both fundraisers and investors. Smart contracts can also ensure that funds are only released when certain conditions are met, which can prevent fraud.
3. **Tokenization:** Blockchain technology can enable the creation of digital tokens that represent ownership of an asset or investment in a project. This can make it easier for investors to participate in crowdfunding campaigns and can also provide liquidity for those tokens.
4. **Transparency and Accountability:** Blockchain technology provides a transparent and immutable record of all transactions, which can increase accountability and trust between fundraisers and investors. Investors can track how their funds are being used, and fundraisers can prove that they are using the funds for the intended purpose.

FUTURE ENHANCEMENT

The future of crowdfunding on blockchain technology looks promising with potential enhancements that can be made to improve its effectiveness and impact. Here are some possible future enhancements:

1. **Integration with AI and Machine Learning:** Blockchain technology can be integrated with AI and machine learning to analyze data and provide more accurate investment recommendations to investors. This can improve the overall quality of investments and increase the likelihood of success for crowdfunding campaigns.
2. **Incentivization Mechanism:** Crowdfunding on blockchain technology can be enhanced by incorporating incentivization mechanisms for both fundraisers and investors. For example, fundraisers can offer rewards to investors who contribute to their campaigns, or investors can earn tokens that appreciate in value over time.
3. **Expansion to new markets:** Crowdfunding on blockchain technology can expand to new markets, such as emerging economies, by providing access to funding and investment opportunities that were previously unavailable. This can stimulate economic growth and create new investment opportunities for investors.
4. **Integration with other blockchain technologies:** Crowdfunding on blockchain technology can be integrated with other blockchain technologies, such as decentralized finance (DeFi) platforms, to create a more comprehensive and interconnected ecosystem for fundraising and investment.

CONCLUSION

Smart Contracts based on Blockchain technology are very suitable to be implemented in 3 dominant crowdfunding process schemes. Beside increasing trust due to blockchain, with smart contract also shorten main process in fundraising. Implementation of Smart Contract based on Blockchain technology, requires high-cost if the organization takes the initiative to implement this technology using their own resources. In general, smart contract service providers use cryptocurrency, where not all governments legally recognize the use of these currencies

REFERENCES

1. Michael Crosby., et.al. BlockChain Technology Beyond Bitcoin, 2015.
2. Simanta Shekhar Sarmah, Understanding Blockchain Technology, 2018.
3. Pauline Adam-Kalfon & Selsabila El Moutaouakil, Blockchain, a catalyst for new approaches in insurance. 5. Satoshi Nakamoto, What is a blockchain.
4. Karim Sultan., et.al, Conceptualizing Blockchains: Characteristics & Applications, 2018.
5. Annisyah Paradhita Sari., et.al, Pengaruh Brand Awareness, Kualitas Proyek Dan Kepercayaan Terhadap Keputusan Berdonasi Secara Online Pada Platform Crowdfunding Kitabisa.Com, 2019.
6. Melfianora, I., & Si, M. (n.d.), Penulisan Karya Tulis Ilmiah Dengan Studi Literatur, 2012.
7. C. D. Clack, V. A. Bakshi, and L. Braine, Smart Contract Templates: foundations, design landscape and research directions, pp. 1–15, 2016, [Online]. Available: <http://arxiv.org/abs/1608.00771>.

Website:

www.google.com

www.youtube.com

www.geeksforgeeks.com

